

SurfaceLayerDiffusivity: first results

First physical comparison | 21 September 2026 | preliminary, diagnostic exclusions apply.

Question: can transport confined to one or two near-wall faces improve coarse LES? Four paired GABLS1 runs use WENO9 at 12.5 m resolution (32^3 cells), a 400 m cube, the same frozen source, and 9 h of integration.

Finding: all three SurfaceLayerDiffusivity variants move the mean u , mean θ and w^2 profiles farther from the fixed 1 m LES median by RMS error over native levels below 200 m. This result applies to this grid and these settings; it is not a general verdict on the closure.

Near-wall response: at 12.5 m, final-hour w^2 falls by 85–93%. Skewness changes from +0.56 in the control to -0.21, -0.25 and -0.42. Domain-integrated resolved TKE falls by 15–20%. The first-face variance is suppressed much more strongly than the integrated energy.

Surface exchange: final-hour u^* rises from 0.279 m/s to 0.292, 0.289 and 0.301 m/s, versus a reference mean of 0.262 m/s. Surface cooling also strengthens, moving farther from the reference. One-face 100 and 300 s responses are similar; two-face support increases local suppression.

Smagorinsky comparison: the earlier 12.5 m WENO9 run ($C_s=0.16$, $Pr=1$) has a final-hour peak w^2 of only $1.12e-6$ m²/s², versus 0.0864 in the control. Its mean- u error is smaller despite almost absent resolved turbulence. Teal curves show this historical context; its source differs from the four matched runs. Log axes reveal the contrast.

Next hypothesis: restrict SLD viscosity to u and v , retaining tracer diffusivity and all other settings. This isolates direct damping of w , but indirect effects through shear and stratification remain. No u/v -only result is included. Single-seed comparisons do not establish uncertainty; the LES median is not observational truth.

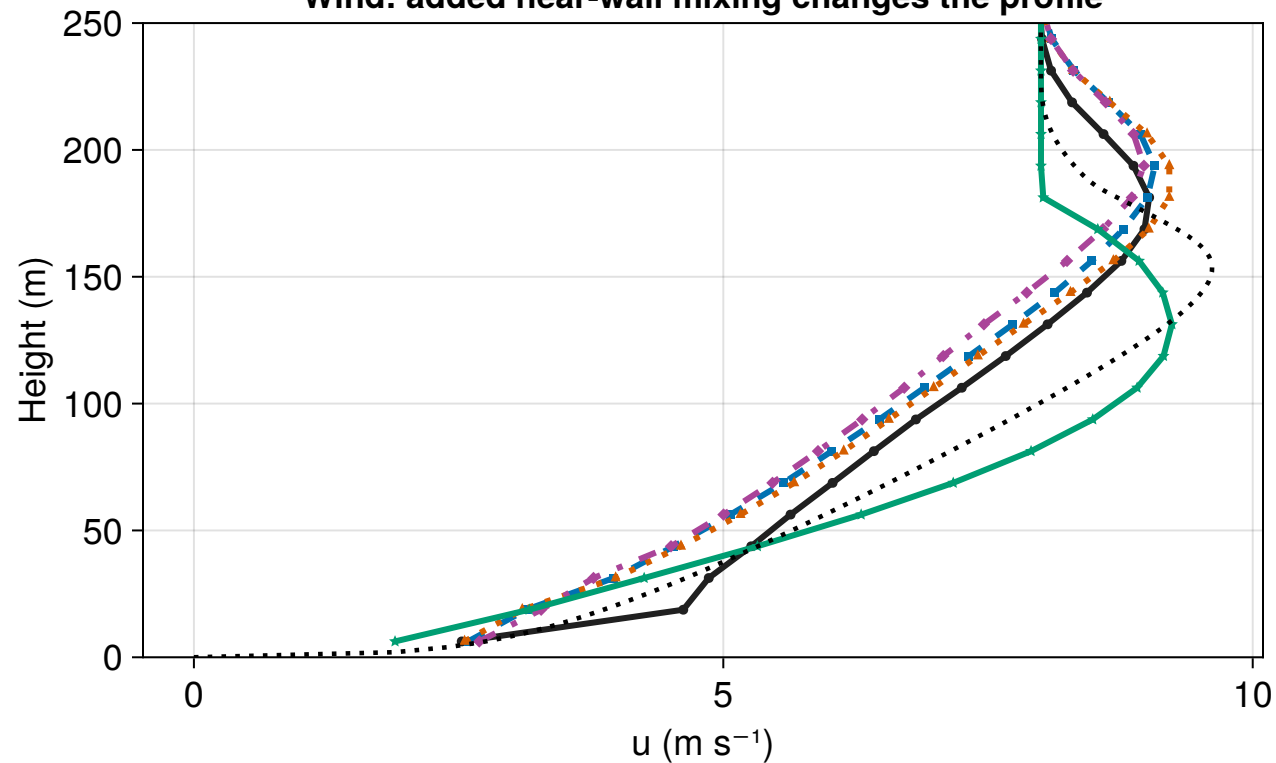
Diagnostic exclusion: SGS flux was incorrectly reported as zero because the generic diagnostic omits the implicit contribution. Combined flux, stress-based depth and dependent budgets are withheld. Mean fields, resolved moments/energy and direct wall exchange remain usable. The model does apply implicit transport. Correct transport diagnostics are the next validation step.

Reading the figures: native-height profiles are true 8–9 h means; timelines use instantaneous output. Colors, line styles and markers distinguish cases. Exact data, Julia source, reference hashes and diagnostic exclusions accompany this brief; all original DYCOMS/GABLS1 report material is retained.

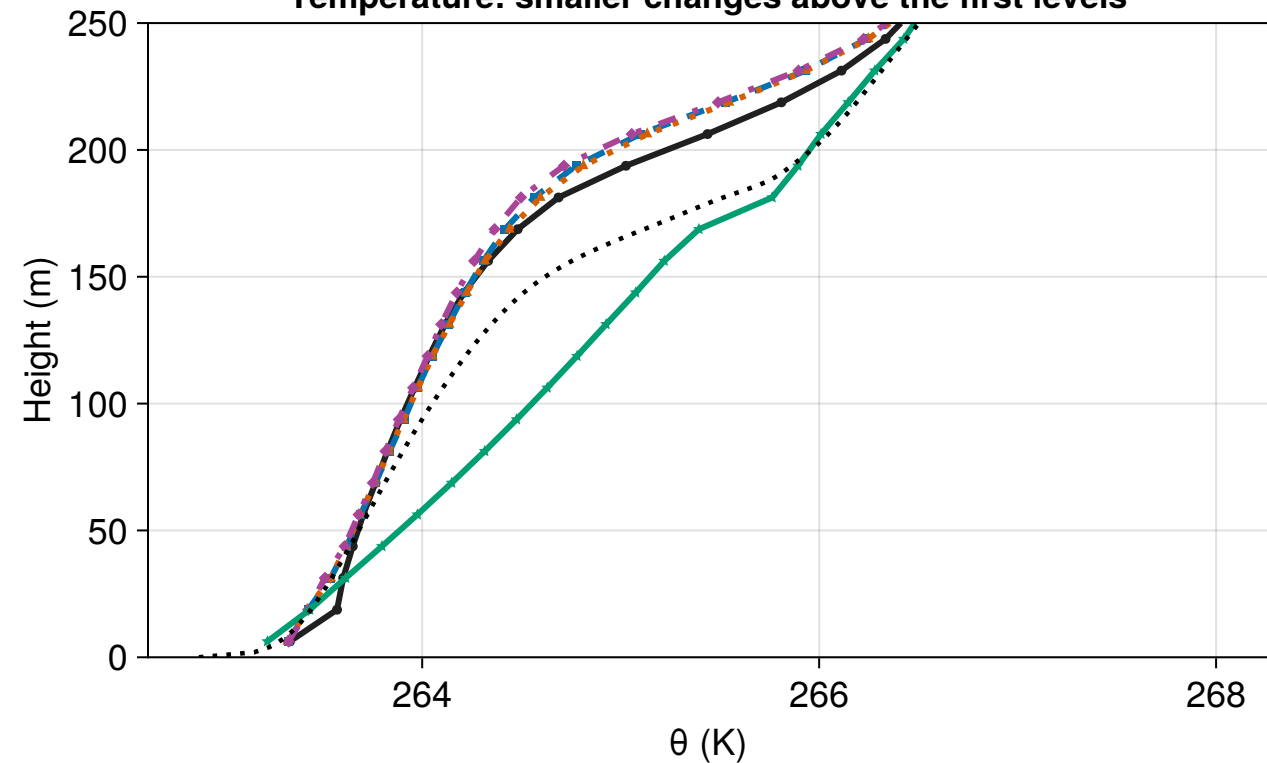
Surface-layer mixing versus coarse-grid Smagorinsky

GABLS1 · WENO9 · 12.5 m grid (32^3) · final hour, 8–9 h · four matched runs + earlier Smagorinsky

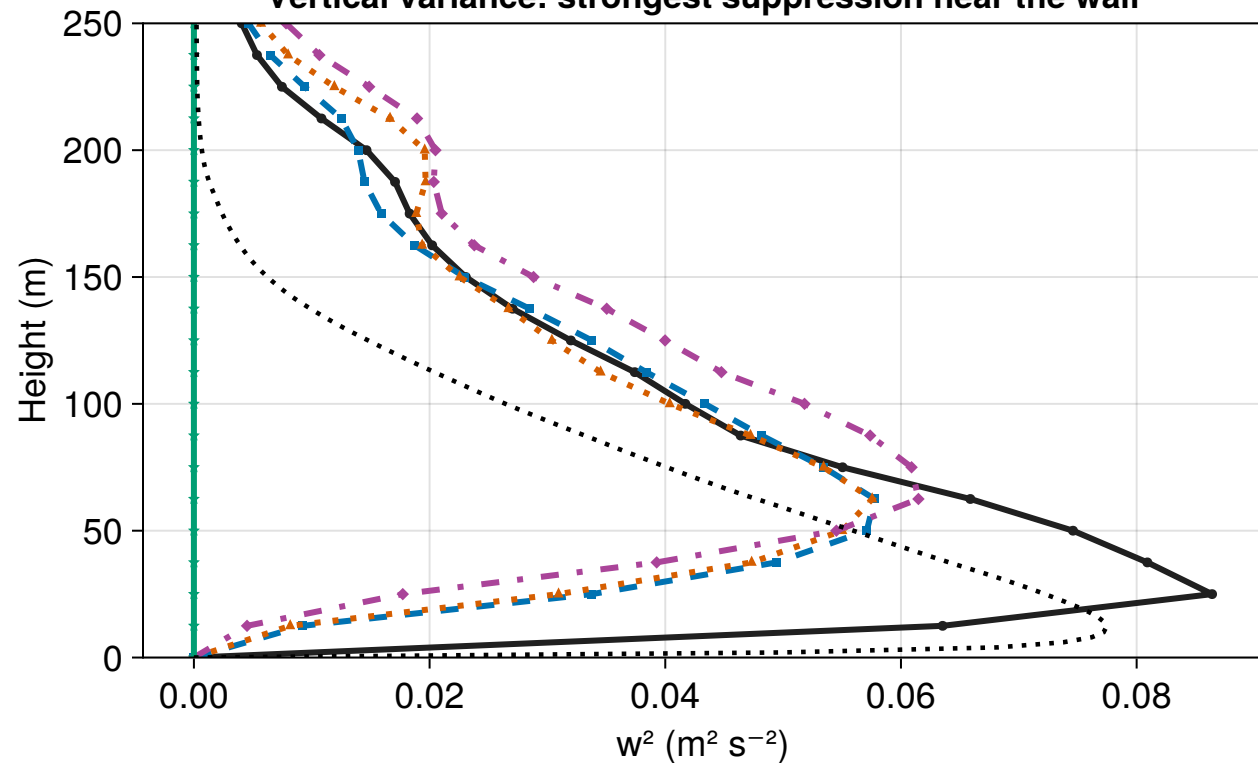
Wind: added near-wall mixing changes the profile



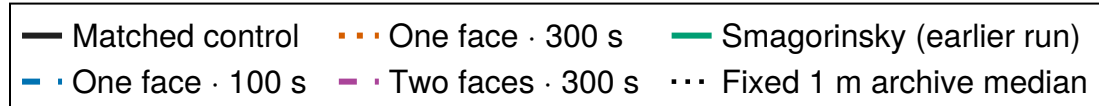
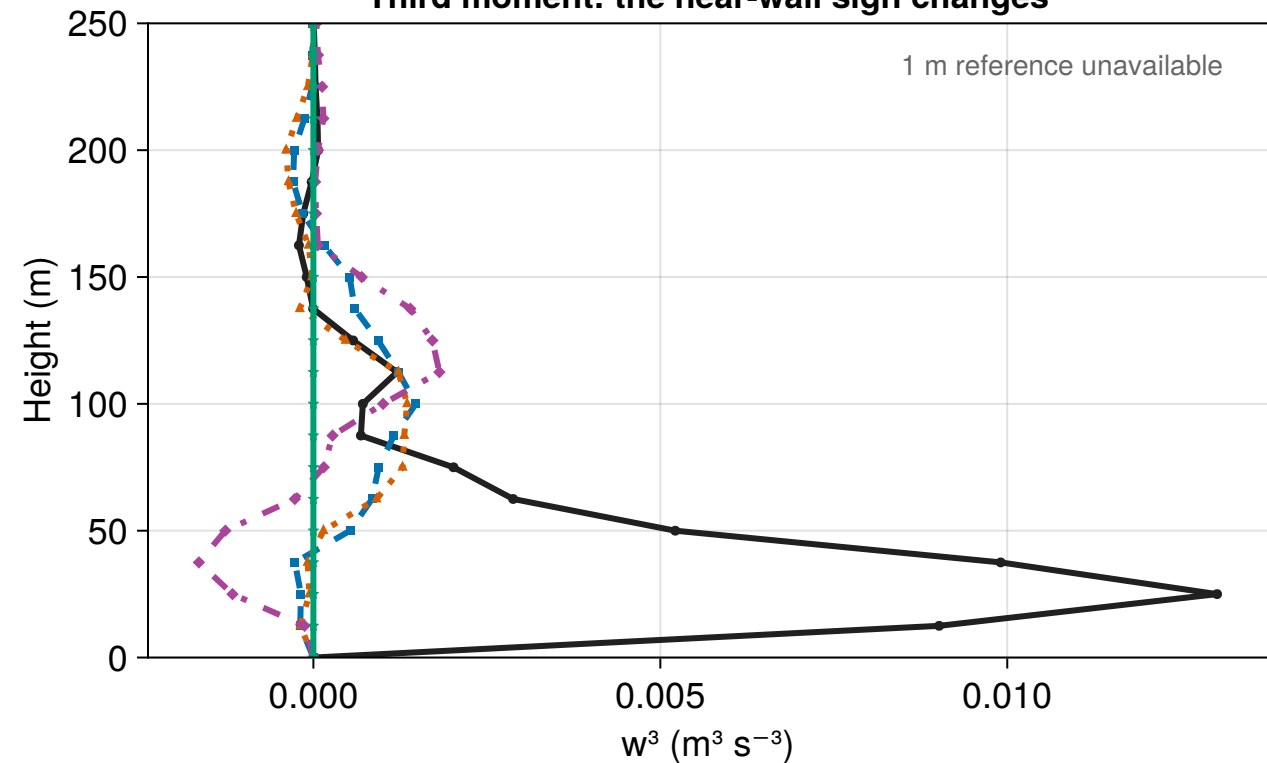
Temperature: smaller changes above the first levels



Vertical variance: strongest suppression near the wall



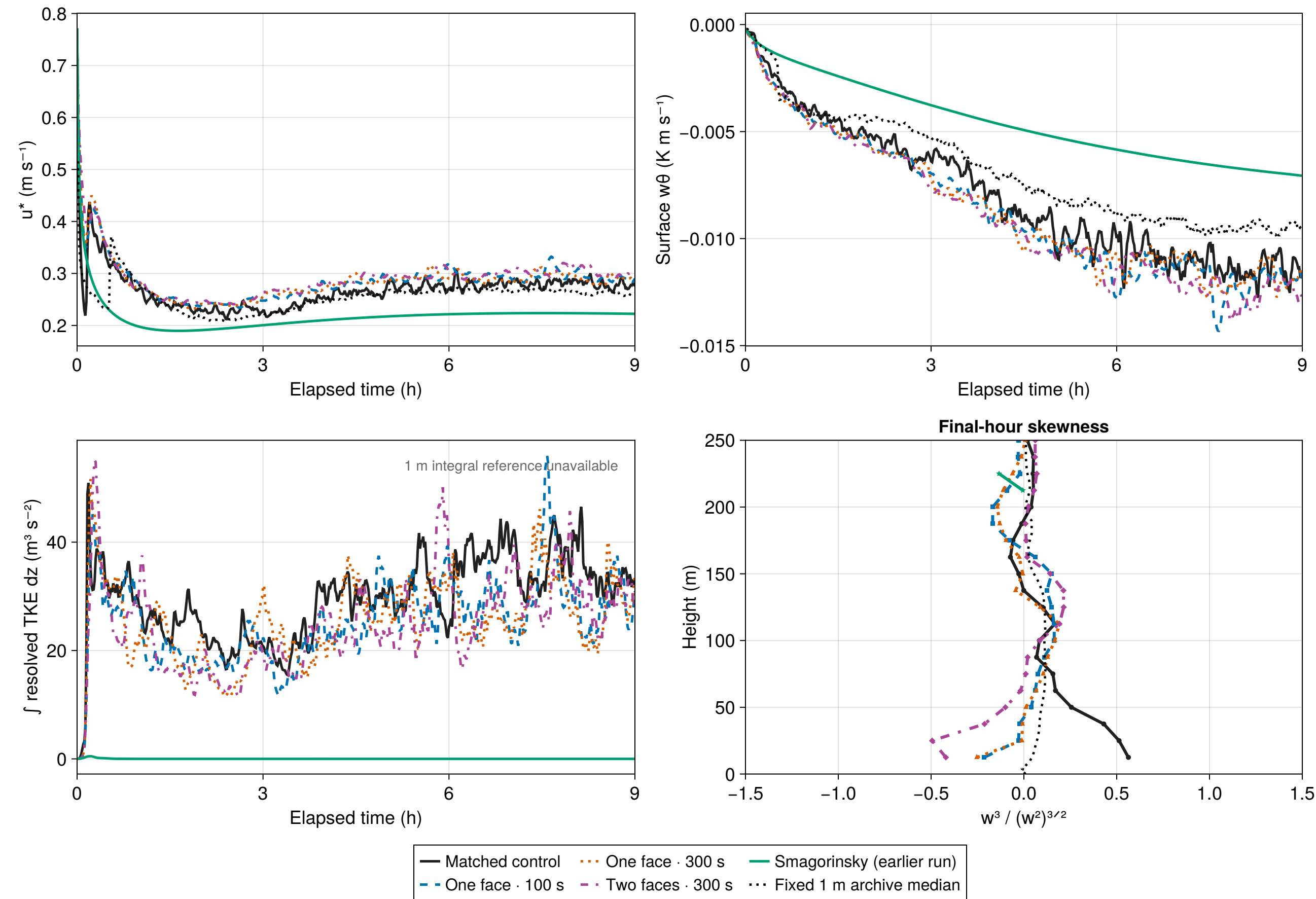
Third moment: the near-wall sign changes



Teal stars: earlier Smagorinsky source; its w^2 and w^3 nearly coincide with zero on these linear axes.
Black dotted: fixed 1 m LES median. SLD SGS/combined flux and dependent diagnostics are withheld.

Follow the response beyond the mean profile

Surface exchange through 9 h; skewness from the final-hour native-face moments

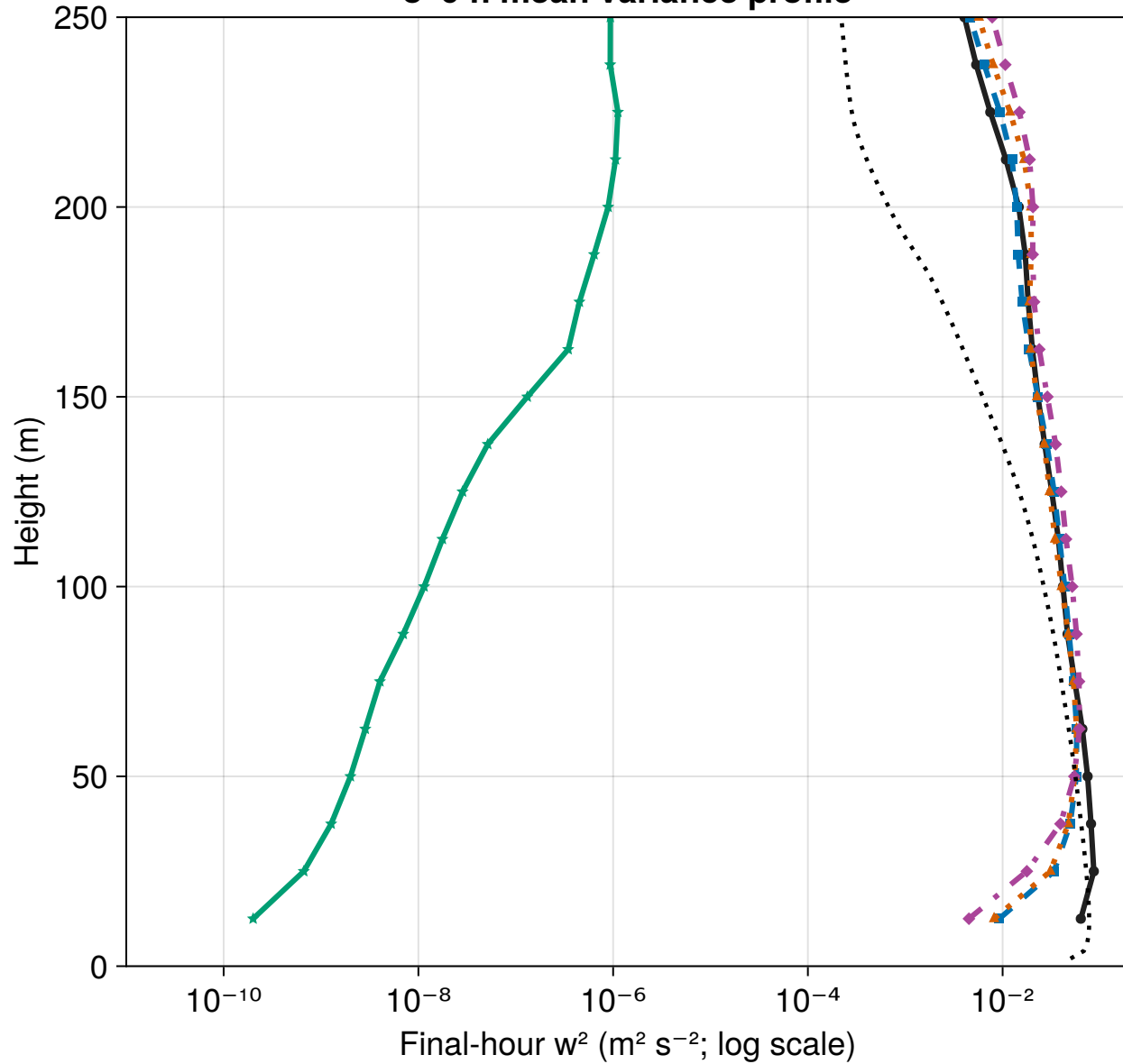


Wall flux remains usable; resolved TKE excludes SGS energy. Teal: earlier Smagorinsky source.
Skewness omits $w^2 < 10^{-6} \text{ m}^2 \text{s}^{-2}$, masking most Smagorinsky levels; a ratio of tiny moments is not robust.

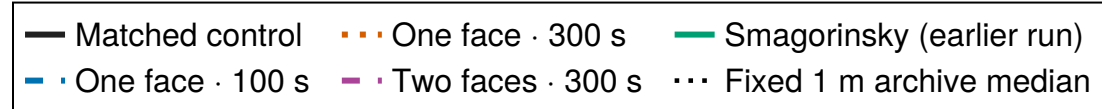
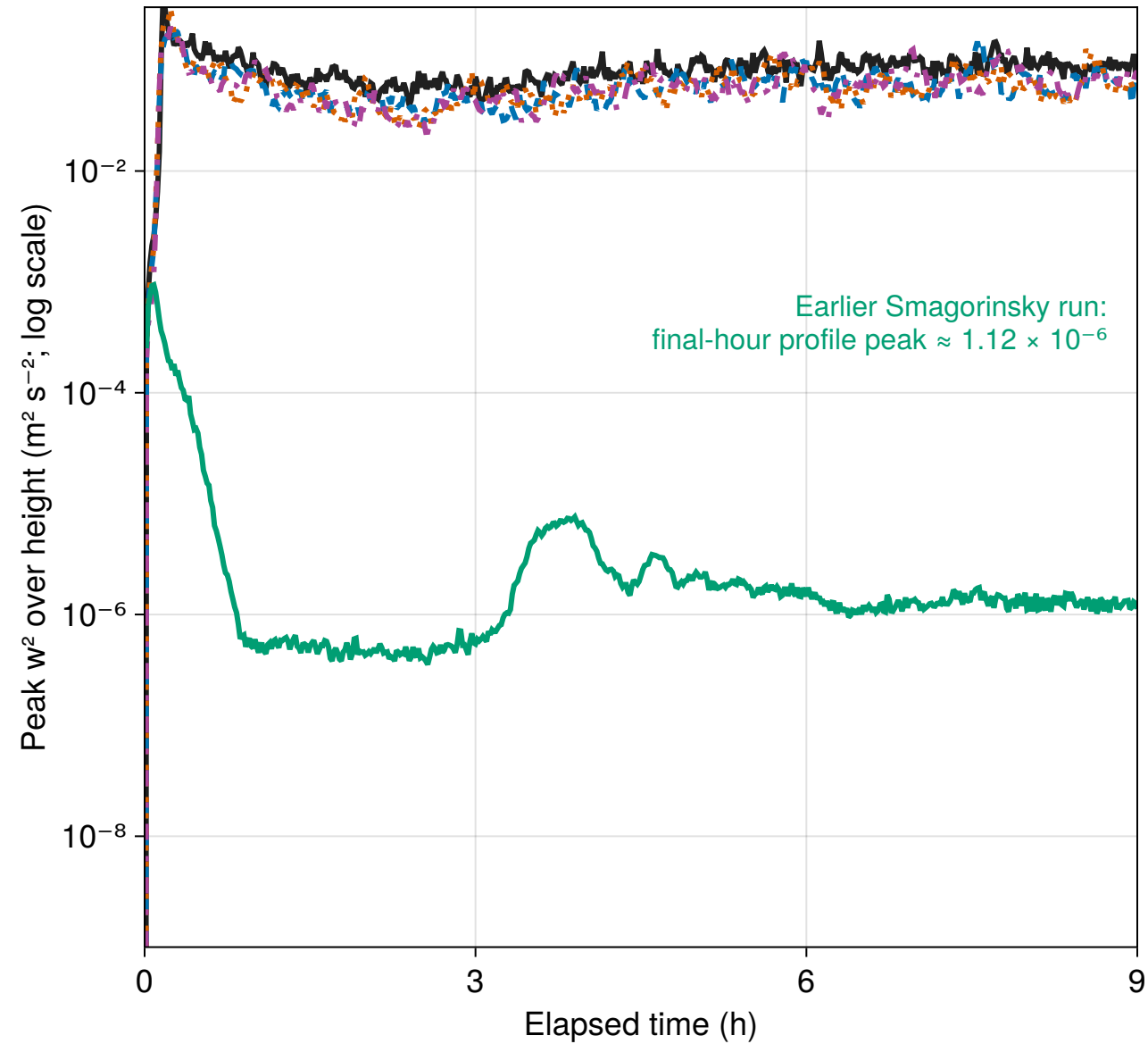
Smagorinsky nearly eliminates resolved vertical turbulence

12.5 m GABLS1 · WENO9 · log axes expose the difference hidden by linear scales

8–9 h mean variance profile



Instantaneous horizontal variance



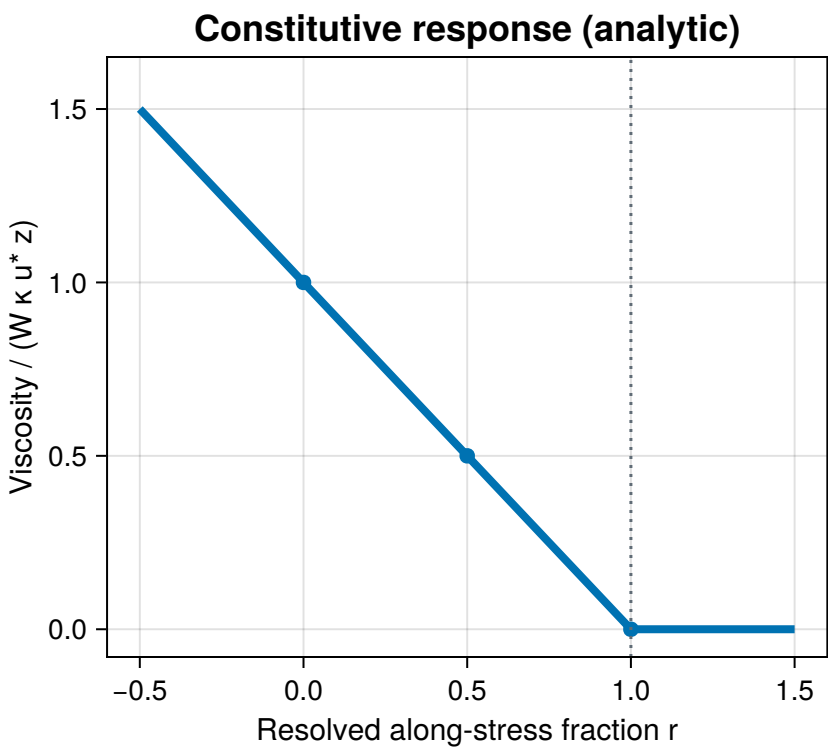
Same grid and advection; Smagorinsky is historical context, not a same-revision controlled treatment.
Zeros are omitted on log axes. The left panel averages first; the right panel takes each instantaneous maximum.

Can a thin surface closure improve coarse LES?

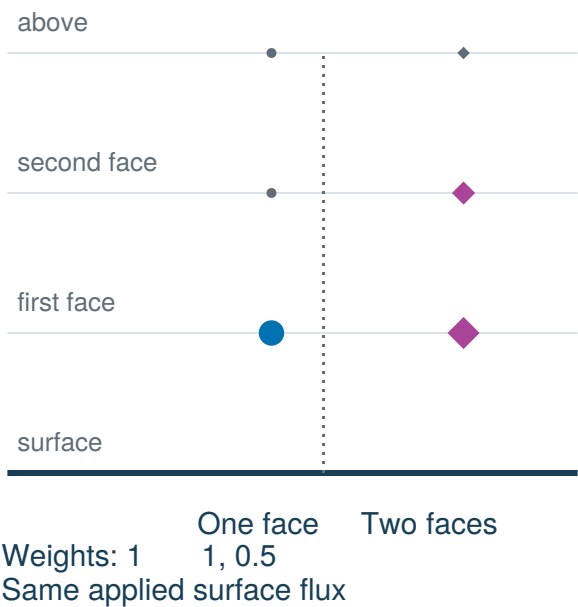
Supply unresolved near-wall transport while preserving resolved turbulence. Four coarse GABLS1 comparisons admitted; transfer tests ongoing.

A local estimate of missing transport

Filter velocity, scalar, and wall-flux samples in physical time. Use their centered covariance to estimate the resolved share of the surface flux.



Shallow vertical support



What this formula does

No resolved stress: full neutral similarity viscosity. Half resolved: half the viscosity. Fully resolved or more: the correction turns off. Opposite-sign transport can give a factor greater than one.

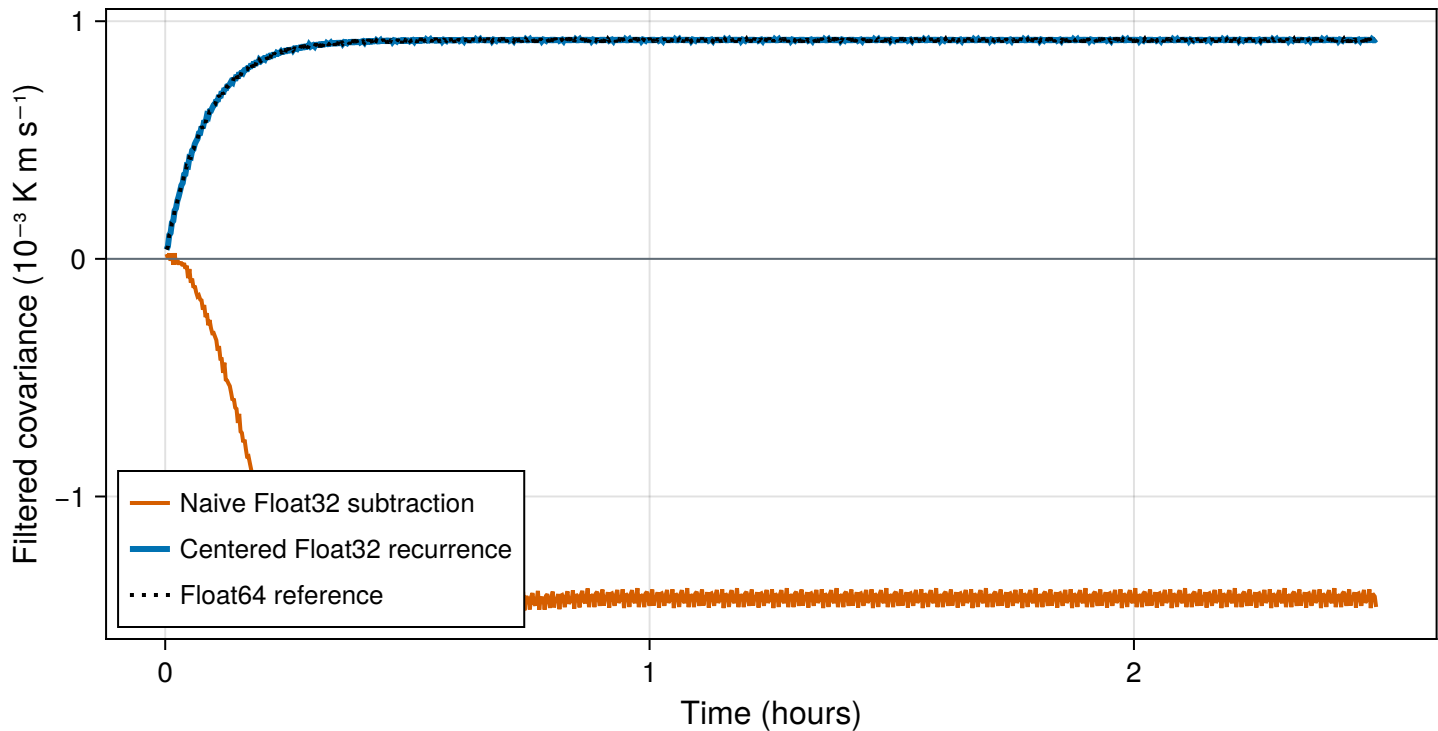
What it does not establish

The added flux still depends on the instantaneous vertical gradient. The formula does not force exact flux balance, include stability functions, or measure the numerical transport produced by WENO.

One-face height: 12.5 m in both planned GABLS comparisons. Scalar correction uses signed flux ratios; zero/tiny wall flux disables it.

A small covariance can get the wrong sign in Float32

Synthetic arithmetic test: 90,000 samples, $\Delta t = 0.1$ s, filter time = 300 s.
Identical Float32 input samples are used in the Float64 reference.



Final covariance: naive -1.465 | centered 0.925 | reference 0.924

Units: $10^{-3} \text{ K m s}^{-1}$. Centered recurrence relative error: 0.020%.

Why this matters for the closure

Subtracting two much larger filtered products loses the small turbulent flux. Here it reverses its sign. A centered covariance update retains the signal that controls the added diffusivity.

Implementation checks support the arithmetic result

The original 119-check CPU suite covers filtering, signs, guards and conservation. Later halo/restart checks and the 1,678-check GPU suite support the admitted run source. Their evidence files retain each revision.

Boundary of the evidence: this arithmetic example is synthetic.
Physical fidelity is assessed separately using the completed GABLS1 runs.
Reproducer, exact plotted CSV, and source hashes accompany this section.

Judge transport, turbulence, and the mean state together

One controlled intervention, three complementary tests of physical behavior. See the accompanying GABLS1 results for the first completed comparison.

01 Did the near-wall transport change as intended?

Lead with resolved, added, and combined momentum/heat flux profiles. Zoom the lowest four levels; mark every active face. Pair flux mismatch with viscosity, diffusivity, guard activity, and mean vertical transport.

02 Did the boundary layer improve without losing turbulence?

Place mean wind and temperature beside w^2 , w^3 and skewness. Show surface flux through time; withhold invalid depth diagnostics. Retain the fixed 1 m GABLS1 median; missing reference quantities remain explicit.

03 Is the result robust to filter time and support?

Use the same four styles in every GABLS panel. Compare with a freshly run, same-revision control, then examine 7-8 h versus 8-9 h. Report changes in physical units; avoid a composite score that hides tradeoffs.

-  Matched WENO9 control
-  One face / 100 s
-  One face / 300 s
-  Two faces / 300 s

Independent challenges strengthen the conclusion

GABLS3: moist nocturnal evolution and the sunrise flux sign change.
Neutral ABL: fixed-stress transport, without a surface-drag prediction.
Both add evidence; neither substitutes for the GABLS1 reference comparison.